

01 Selecting and Sorting.

01: Retrieve all columns from the employee table.

SQL Query

SELECT *

FROM employees;

ID	First Name	Last Name	Dept	Salary	hire Date	City
1	Alice	Green	IT	70000	20-01-10	JHB
2	Brian	Lee	HR	45000	19-03-22	CT
3	Cathy	Zeller	Fin	65000	18-07-18	DBT
4	David	Mulora	Market	50000	21-11-05	PTA
5	Eva	Maidoo	IT	72000	17-09-30	JHB

02 Find all unique departments

SQL Query

SELECT DISTINCT department
FROM employees

Department
IT
HR
Fin
Marketing

03 Retrieve first and last names ordered by salary descending.

SQL Query

```
SELECT first_name, last_name, salary  
FROM employees  
ORDER BY salary DESC;
```

First name	Last name	Salary
Eva	Maidoo	72 000
Alice	Green	70 000
Cathy	Zeller	65 000
David	Mokoena	50 000
Brian	Lee	45 000

04 Retrieve the top 3 highest-paid employees.

```
SELECT id, first_name, last_name, salary  
FROM employees  
ORDER BY salary DESC  
LIMIT 3;
```

ID	first name	Last name	Salary
5	Eva	Maidoo	72 000
1	Alice	Green	70 000
3	Cathy	Zeller	65 000

Q2 Filtering with WHERE.

Q5 Find employees in IT department.

```
SELECT id, first_name, last_name, department
FROM employees
WHERE department = 'IT';
```

ID	first_name	last_name	Dept
1	Alic	Green	IT
5	Sam	Plaidoo	IT

Q6 Find employees in finance with salary > 60000

```
SELECT id, first_name, last_name, department, salary
FROM employees
WHERE department = 'Finance' AND salary > 60000;
```

ID	first_name	last_name	Dept	Salary
3	Cheryl	Zaki	fin	65000

Q7 Find employees in HR or Marketing.

```
SELECT id, first_name, last_name, department
FROM employees
WHERE department = 'HR' OR
      department = 'Marketing';
```

ID	first_name	last_name	Department
2	Brian	Lee	HR
4	David	Mohanna	Marketing

08 Find employees not in IT.

```
SELECT id, first_name, last_name, department
FROM employees
WHERE NOT department = 'IT'
```

ID	first_name	last_name	Dept
2	Brian	Lee	HR
3	Gary	Zak	Fin
4	Drew	Moham	Marketing

09 Find employees in IT, HR, or Finance using IN

```
SELECT id, first_name, last_name, department
FROM employees
WHERE department IN ('IT', 'HR', 'Finance')
```

ID	first_name	last_name	Dept.
1	Alice	Green	IT.
2	Brian	Lee	HR
3	Gary	Zak	Finance
4	Eva	Naidu	IT

10 Find employees in IT with Salary > 65000 and city Johannesburg.

SELECT id, first_name, last_name, department,
Salary, city

FROM employees

WHERE department = 'IT'

AND salary > 60000

AND city = 'Johannesburg'

ID	first_name	last_name	Dept	Salary	City
1	ALB	Green	IT	70000	JHB
5	SLV	Plaidoo	IT	72000	JHB